Chapter 4 exercises

DS04-E01 • The last queue node

A linked queue stores head and tail. From empty, enqueue 4, dequeue, then enqueue 9. Draw pointers and ownership at every state. Explain the failure if removing the last node updates only head.

DS04-E02 • Proving in-place reversal

Reverse `3→8→11→null` using three pointers. List the lists reached by prev and curr after every iteration. State the disjointness and coverage invariant. How must tail change if it is also stored?

DS04-E03 • Polynomial normalization and addition

P has input terms `(4,2),(1,1),(4,-1)`. Q has `(4,-1),(0,5)`. Explain why P is not canonical. Find canonical P+Q, and analyze sorting and merging separately.

DS04-E04 • Doubly linked deletion

A doubly linked list has head and tail sentinels, and x is a known data node. Give O(1) deletion steps and prove consistency of the remaining neighbors. State invalid x or sentinel states that the API contract must exclude.

DS04-E05 • Last node of a circular queue

A circular queue keeps only tail. Enqueue A, enqueue B, dequeue twice, then enqueue C. Draw the links and tail after every operation. Why must the second dequeue not retain tail after releasing the node?

DS04-E06 • Fast and slow pointers and the entry

For `0→1→2→3→4→2`, trace slow and fast through the first meeting and then both pointers after resetting one to head. Give μ, λ, and the entry. What must be compared if distinct nodes hold equal values?

DS04-E07 • Acyclic and self-loop boundaries

Run Floyd's algorithm on an empty chain, `A→null`, and `A→A`. State where the fast-pointer check stops or when the pointers meet, and explain why each pointer read is valid.

DS04-E08 • Insertion order and reachability

The chain is `3→8→11→null`, p points to 3, and new node x holds 5. If `p.next=x` executes before `x.next=p.next`, what can be reached from head? Give the correct update order and explain the effect on ownership of the old suffix.

DS04-E09 • A cycle beginning at head

Two distinct nodes A and B both hold 7, with links `A→B→A` and head=A. Trace Floyd's algorithm, giving μ, λ, its first meeting, and the entry. Why would comparing the two node values after the first step be wrong?

DS04-E10 • The only data node between sentinels

A doubly linked list is `H↔x↔T`, with non-data sentinels H and T. Which fields change when x is deleted? Give the resulting size and sentinel links, and say whether an external pointer to x may be dereferenced.

DS04-E11 • Copying and node ownership

A linked-list class owns head through a raw pointer but uses compiler-generated shallow copy. What happens when both copies are destroyed? Give deep-copy and noncopyable alternatives, and explain why even a non-owning tail must be updated after deletion.

References and attribution

Lin Cing, handwritten data structures notes; C++ draft forward_list, list and unique_ptr.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.
https://github.com/xinhi6716/computer-science-self-study